

A Cancellation-Safe Derivative Lower Bound for the Exact- U_c Polar Fredholm Determinant

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Abstract

For the unchanged canonical Fredholm determinant of the frozen exact- U_c polar Logistic transfer family, we prove a cancellation-safe lower-growth statement at the real point $s = 2$. The exact based-word trace logarithm is absolutely convergent there, and every signed denominator remains positive on the real axis. An absolute trace majorant bounds the determinant itself from below, while the pure-left length-one orbit gives a positive term in its logarithmic derivative. Writing

$$\alpha_0 = \frac{U_c^2}{4}, \quad \tau_* = -\log \alpha_0, \quad B_2 = \frac{-\log(1 - 2\alpha_0^2)}{1 - \alpha_0},$$

we obtain

$$D'_{\text{pol}}(2) \geq e^{-B_2} \frac{\tau_* \alpha_0^2}{1 - \alpha_0} > 0.0213.$$

Cauchy's estimate therefore yields $M_D(R) > 0.0213(R - 2)$ for $R > 2$, and $M_D(R) > 0.01065R$ for $R \geq 4$. Moreover, $D_{\text{pol}}(\sigma) \rightarrow 1$ as $\sigma \rightarrow +\infty$; the nonzero derivative then proves that D_{pol} is transcendental entire, so its maximum modulus grows faster than every fixed power. These conclusions do not prove positive or exact order, an exponential lower bound, a zero-count lower bound, a $T \log T$ law, completed-xi structure, or the Riemann hypothesis.

1 Frozen determinant and main result

Let $u = U_c$ be the real root of

$$u^3 - 2u^2 + 2u - 2 = 0, \quad \rho = u - 1. \tag{1}$$

The inherited real system is the exact two-full-branch polar map on $I_L = [-\pi/2, 0]$ and $I_R = [0, \pi/2]$. Its inverse branches ϕ_L, ϕ_R extend to the fixed complex stadiums and share a holomorphic logarithmic weight $\ell = \text{Log } a$, where $\phi'_L = a$ and $\phi'_R = -a$.

On the matching space

$$X = A(U_L) \oplus A(U_R), \quad B = \{(v_L, v_R) : v_L(0) = v_R(0)\},$$

the transfer family is

$$(\mathcal{L}_s v)_j(z) = e^{s\ell(z)} [v_L(\phi_L(z)) + v_R(\phi_R(z))]. \tag{2}$$

The preceding analytic stages establish that $\mathcal{L}_{s,B} := \mathcal{L}_s|_B$ is nuclear of order zero and that

$$D_{\text{pol}}(s) := \det_{\text{Fr}}(I - \mathcal{L}_{s,B}) \tag{3}$$

is entire. Equation (3), the roof clock, and the signed trace convention are fixed throughout this paper.

Define

$$\alpha_0 = \frac{U_c^2}{4}, \quad \tau_* = -\log \alpha_0, \quad B_2 = \frac{-\log(1 - 2\alpha_0^2)}{1 - \alpha_0}, \quad (4)$$

and let

$$M_D(R) = \max_{|s| \leq R} |D_{\text{pol}}(s)|. \quad (5)$$

Theorem 1 (Derivative and lower growth). *For the same determinant (3),*

$$D'_{\text{pol}}(2) \geq e^{-B_2} \frac{\tau_* \alpha_0^2}{1 - \alpha_0} > 0.0213. \quad (6)$$

Consequently,

$$M_D(R) > 0.0213(R - 2) \quad (R > 2), \quad (7)$$

and

$$M_D(R) > 0.01065R \quad (R \geq 4). \quad (8)$$

The function D_{pol} is transcendental entire. In particular, for every fixed $A > 0$,

$$\lim_{R \rightarrow \infty} \frac{M_D(R)}{R^A} = +\infty. \quad (9)$$

The explicit inequality in (7) is only linear. The qualitative statement (9) does not supply an effective coefficient sequence, an exponential rate, or a positive entire-function order.

2 Signed trace logarithm on the safe real axis

For a based word $\omega \in \{L, R\}^n$, let $T_\omega > 0$ denote its roof time and retain the exact orientation sign

$$\varepsilon_\omega = (-1)^{\#R(\omega)}. \quad (10)$$

The inherited trace identity is

$$\text{Tr } \mathcal{L}_s^n = \sum_{\omega \in \{L, R\}^n} \frac{e^{-sT_\omega}}{1 - \varepsilon_\omega e^{-T_\omega}}. \quad (11)$$

No denominator in (11) is replaced by an unsigned expression.

The real roof bound gives

$$T_\omega \geq n\tau_*, \quad e^{-T_\omega} \leq \alpha_0^n. \quad (12)$$

For $\sigma = \Re s \geq 0$, equations (11)–(12) imply the absolute majorant

$$|\text{Tr } \mathcal{L}_s^n| \leq \frac{(2\alpha_0^\sigma)^n}{1 - \alpha_0^n}. \quad (13)$$

Thus, when

$$\sigma > \sigma_* := \frac{\log 2}{-\log \alpha_0}, \quad q_\sigma := 2\alpha_0^\sigma < 1,$$

the series converges normally and

$$\sum_{n \geq 1} \frac{|\operatorname{Tr} \mathcal{L}_s^n|}{n} \leq \frac{-\log(1 - q_\sigma)}{1 - \alpha_0} =: B(\sigma). \quad (14)$$

Keeping the auxiliary Fredholm variable distinct from s , analytic continuation from its convergence disk gives

$$D_{\text{pol}}(s) = \exp\left(-\sum_{n \geq 1} \frac{\operatorname{Tr} \mathcal{L}_s^n}{n}\right) \quad (\Re s > \sigma_*). \quad (15)$$

For real $\sigma > \sigma_*$, both possible denominators in (11) are strictly positive. If $S(\sigma)$ denotes the complete positive trace sum in the exponent of (15), then

$$S(\sigma) > 0, \quad \log D_{\text{pol}}(\sigma) = -S(\sigma) < 0.$$

Consequently,

$$0 < \exp[-B(\sigma)] \leq D_{\text{pol}}(\sigma) \leq 1. \quad (16)$$

At $\sigma = 2$, equation (14) is exactly the constant B_2 in (4).

3 The pure-left derivative lower bound

Normal convergence of the trace logarithm on closed sub-half-planes permits termwise differentiation. For real $\sigma > \sigma_*$, equation (15) gives

$$\frac{d}{d\sigma} \log D_{\text{pol}}(\sigma) = \sum_{n \geq 1} \frac{1}{n} \sum_{\omega \in \{L, R\}^n} \frac{T_\omega e^{-\sigma T_\omega}}{1 - \varepsilon_\omega e^{-T_\omega}}. \quad (17)$$

Every summand in (17) is positive. This use of positivity is cancellation-safe: it occurs only after the exact signed denominator has been fixed, and no complex or signed sum is split into incompatible absolute bounds.

The boundary periodic point $P = -\pi/2$ has one left label. For the length-one pure-left word,

$$T_L = \tau_*, \quad e^{-T_L} = \alpha_0, \quad \varepsilon_L = 1. \quad (18)$$

Retaining this single positive term in (17) at $\sigma = 2$ yields

$$\left. \frac{d}{d\sigma} \log D_{\text{pol}}(\sigma) \right|_{\sigma=2} \geq \frac{\tau_* \alpha_0^2}{1 - \alpha_0}. \quad (19)$$

Meanwhile, (16) gives $D_{\text{pol}}(2) \geq e^{-B_2}$. Since $D'_{\text{pol}}(2) = D_{\text{pol}}(2)(\log D_{\text{pol}})'(2)$, multiplying the two positive lower bounds proves

$$D'_{\text{pol}}(2) \geq e^{-B_2} \frac{\tau_* \alpha_0^2}{1 - \alpha_0}. \quad (20)$$

This mechanism avoids the usual lower-bound obstruction from cancellation. It works at a real point where the exact orbit ledger is termwise positive; it does not assert positivity for complex s , nor does it transfer a single coefficient through an uncontrolled signed determinant expansion.

4 Maximum modulus and transcendence

Fix $R > 2$. The closed disk centered at 2 with radius $R - 2$ is contained in $\{|s| \leq R\}$. Cauchy's derivative estimate on that disk therefore gives

$$|D'_{\text{pol}}(2)| \leq \frac{M_D(R)}{R - 2}. \quad (21)$$

Combining (20) with its certified decimal lower bound proves (7). If $R \geq 4$, then $R - 2 \geq R/2$, which proves (8).

The trace logarithm supplies a separate qualitative conclusion. Since $q_\sigma = 2\alpha_0^\sigma \rightarrow 0$, equation (14) gives $B(\sigma) \rightarrow 0$. Equations (15)–(16) therefore imply

$$D_{\text{pol}}(\sigma) \rightarrow 1 \quad (\sigma \rightarrow +\infty). \quad (22)$$

The derivative bound (20) proves that D_{pol} is nonconstant. A nonconstant polynomial diverges in modulus along the positive real ray, so (22) rules out every polynomial. Because D_{pol} is entire, it is transcendental entire.

Finally, write $D_{\text{pol}}(s) = \sum_{m \geq 0} a_m s^m$. Transcendence means that nonzero coefficients occur at arbitrarily high degrees. For any fixed $A > 0$, choose an integer $m > A$ with $a_m \neq 0$. Cauchy's coefficient estimate gives

$$M_D(R) \geq |a_m| R^m,$$

and hence

$$\frac{M_D(R)}{R^A} \geq |a_m| R^{m-A} \rightarrow \infty.$$

This proves (9), but it gives no useful numerical value for $|a_m|$ as m varies.

5 Outward interval certificate and reproducibility

The companion scalar audit evaluates (1) and (4) with 1024-bit outward Arb balls. Table 1 displays central decimal values; the machine-readable artifact retains the interval radii and the strict comparison gates.

Quantity	Certified displayed value
U_c	1.54368901269207636157085597180...
α_0	0.59574394197655937353067713411...
τ_*	0.51794433178925625819844058117...
$2\alpha_0^2$	0.70982168880354028313796314116...
B_2	3.06058413770995492489591306195...
e^{-B_2}	0.04686031434695642308253249628...
$\tau_* \alpha_0^2 / (1 - \alpha_0)$	0.45472184398972339295137217650...
Lower-bound product	0.02130840854978611545501449300...

Table 1: Target-free scalar evaluation. The final outward interval lies strictly above 0.0213; no Fredholm determinant value is evaluated.

The certificate checks that the algebraic root lies in its sealed bracket, $0 < \alpha_0 < 1$, $2\alpha_0^2 < 1$, $2 > \sigma_*$, every logarithm has a positive argument, and the final outward lower endpoint exceeds 0.0213. Its data firewall forbids prime tables, primality predicates, Riemann-zero tables, zeta or xi evaluation, USTC data, parameter fitting, Fredholm evaluation, and root searches.

The source, regression, and artifact are synchronized as one provenance unit under `src/`, `tests/`, and `artifacts/log_0001_lower_growth/`. The scalar program certifies only the numerical inequalities used after the analytic trace argument; it cannot certify termwise positivity or the Cauchy step by sampling.

6 Limitations and conclusion

The safe real axis supplies a narrow way around signed cancellation. At $s = 2$, the unchanged trace logarithm converges absolutely, all exact real denominators are positive, and the pure-left orbit gives a retained positive term. This proves $D'_{\text{pol}}(2) > 0.0213$ without evaluating the determinant. Cauchy's estimate transfers the local derivative certificate to an explicit linear lower bound for the global maximum modulus.

The limiting value $D_{\text{pol}}(\sigma) \rightarrow 1$ and the nonzero derivative also prove that the determinant is transcendental entire. The resulting super-polynomial maximum-modulus statement is qualitative. It does not control any high Taylor coefficient uniformly and therefore cannot be promoted to an exponential lower bound or a positive-order theorem.

The claim boundary is strict. This stage does not prove positive order, exact order, finite nonzero type, an exponential lower bound, a zero-count lower bound, a sharp divisor asymptotic, or a $T \log T$ law. It does not identify determinant zeros, relate primitive roof periods to logarithms of primes, produce von-Mangoldt weights, establish a completed-xi determinant, or construct a self-adjoint operator. Route B remains unauthorized, and no statement about Hilbert–Polya or the Riemann hypothesis follows.

The reusable conclusion is structural: a same-object trace logarithm can produce a certified determinant lower statement when one finds a real zero-free point where its exact signed terms become positive. Beyond that pointwise mechanism, quantitative lower growth still requires new information about high coefficients, boundary values, or the determinant divisor.

A Proof details

A.1 Normal convergence and differentiation

Fix a compact set $K \subset \{s : \Re s > \sigma_*\}$, and let $\sigma_0 = \min_{s \in K} \Re s$. Equation (13) gives

$$\sum_{n \geq 1} \sup_{s \in K} \frac{|\text{Tr } \mathcal{L}_s^n|}{n} \leq \sum_{n \geq 1} \frac{(2\alpha_0^{\sigma_0})^n}{n(1 - \alpha_0^n)} < \infty.$$

Thus the trace logarithm is a normally convergent series of holomorphic functions. Cauchy's integral formula on a slightly larger compact neighborhood permits termwise differentiation, which proves (17). For real σ , $0 < e^{-T_\omega} < 1$ and $\varepsilon_\omega \in \{-1, 1\}$, so every denominator and every numerator in (17) is positive.

A.2 The determinant lower anchor

For real $\sigma > \sigma_*$, put

$$S(\sigma) = \sum_{n \geq 1} \frac{1}{n} \sum_{\omega \in \{L, R\}^n} \frac{e^{-\sigma T_\omega}}{1 - \varepsilon_\omega e^{-T_\omega}}.$$

Termwise positivity and (14) give $0 < S(\sigma) \leq B(\sigma)$. Equation (15) becomes $D_{\text{pol}}(\sigma) = e^{-S(\sigma)}$, hence

$$e^{-B(\sigma)} \leq D_{\text{pol}}(\sigma) < 1.$$

At $\sigma = 2$, $q_2 = 2\alpha_0^2$, so $B(2) = B_2$. Combining this anchor with the pure-left term in (17) proves (20).

A.3 Cauchy transfer

For $0 < r < R - 2$, Cauchy's estimate on $|s - 2| \leq r$ gives

$$|D'_{\text{pol}}(2)| \leq \frac{\max_{|s-2| \leq r} |D_{\text{pol}}(s)|}{r} \leq \frac{M_D(R)}{r}.$$

Letting $r \uparrow R - 2$ proves (21). This centered-disk argument avoids the extraneous factor that would arise from estimating the derivative directly on the circle $|s| = R$.

A.4 Super-polynomial maximum modulus

If an entire function $f(s) = \sum_{m \geq 0} a_m s^m$ is not a polynomial, then for every $A > 0$ there is an integer $m > A$ with $a_m \neq 0$. Cauchy's coefficient estimate on $|s| = R$ gives $|a_m| \leq M_f(R) R^{-m}$. Therefore $M_f(R)/R^A \geq |a_m| R^{m-A} \rightarrow \infty$. Applying this observation to $f = D_{\text{pol}}$ completes the proof of (9).

References

No external references are cited in this self-contained stage report. The argument uses the frozen determinant and trace ledger established in the preceding project stages and standard facts from complex analysis.